Integration by Parts

The integration by parts formula

1 Evaluate $\int xe^x\,dx$ using integration by parts.

2 Evaluate $\int x\cos x\,dx$ using integration by parts.

3 Evaluate $\int \ln x\,dx$ using integration by parts (with $dv = dx$).

Repeated and definite applications

4 Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.

5 Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.

Choosing u and dv wisely (LIATE)

6 Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).

7 Evaluate $\int x^2\cos x\,dx$, applying integration by parts twice.

The recurring-integral trick

8 Evaluate $\int e^x\sin x\,dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

Definite integrals by parts

9 Evaluate $\int_0^\pi x\sin x\,dx$ exactly.

10 Evaluate $\int_1^e x^2\ln x\,dx$ exactly.

Reduction formulas

11 Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1} e^x\,dx$.

12 Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x\,dx$, confirming it matches an earlier result.

Integration by parts with inverse trig functions

13 Evaluate $\int \tan^{-1} x\,dx$ using integration by parts with $u = \tan^{-1} x$, $dv = dx$.

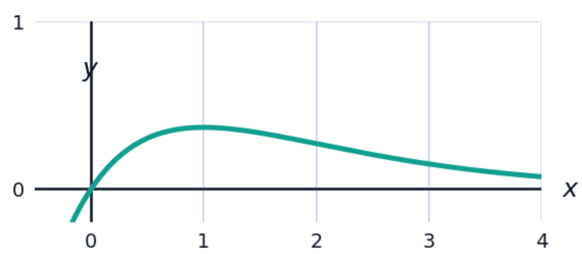
14 Evaluate $\int x\sin^{-1} x\,dx$ using $u = \sin^{-1} x$, $dv = x\,dx$ (the resulting integral can be left in terms of a known antiderivative form).

A capstone mixed-technique problem

15 Evaluate $\int_0^{\pi/2} x\cos(2x)\,dx$ exactly, using integration by parts.

Visualizing an area found by integration by parts

16 The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.



Choosing between substitution and parts

17 For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u-substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

18 Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

19 A city's population growth rate is modeled by $r(t) = t\ln(t + 1)$ thousand people per year, for $0 \leq t \leq 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.